1. DETERMINATION OF BLOOD GROUPING

Aim

To determine the blood group of the given blood sample.

Principle

This test is based on the antigen antibody complex formation between the antigen present on the RBC and the antibody present in the serum.

Reagents and Equipments

1. Anti A, 2. Anti B, 3. Anti D, 4. Sterile lancet, 5. White marble tiles, 6. Sticks for mixing and 7. Alcohol

Procedure

The middle finger of the individual to be tested for blood group is cleaned with alcohol and the excess of alcohol is wiped out with sterile cotton. A clean white tile is taken and it is divided into four columns marked as A, B, D and C.

A small drop of anti A is added in the portion A, anti B in the portion B and anti D in the portion D. The portion C is used for positive or negative control. A small prick is made on the cleaned finger, the first drop of blood coming out is wiped off and second drop of blood is collected directly on the region marked as A, B and D. Immediately, blood is mixed with the corresponding antibody by using the sterile stick and observed for any agglutination in the form of clump formation.

If the clump is observed in the region marked as A, the blood group is A and the formation of clump in B shows presence of B blood group. If the clump is formed in both A and B , the blood group is of AB type. If there is no clump formation in both A and B, the blood group, is of O type. The clump formation in D portion is observed carefully. The formation of clump in the D region shows presence of Rh positive blood group and if there is no clump formation it shows the presence of Rh negative blood group.

Clump formation in the regions A and D shows the presence of A+ blood group and clump formation in the regions B and D shows the presence of B+ blood group. If clump is formed in all the three regions it shows the presence of AB+ blood group. No clump formation in regions A & B and clump formation in region D shows O+ blood group. No clump formation in all the three regions shows the presence of O- blood group. The following chart shows the type of blood group and the agglutination with anti-bodies.

If the clump formation is observed immediately in A blood group type it can be denoted as A_1 type, if it is not immediate then the blood group is of A_2 type.

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